

Stage 3I Physics Note: Small- x Regularity for Cartoon Variables

GL-with-AI Project

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1 Scope

Stage 3I documents regularity requirements near the cartoon/spherical axis $x = 0$ for the future $4 + 1$ black-string modified-cartoon implementation. It is a documentation and symbolic-limit note only. It does not implement C++ source terms, finite differences, ghost-cell filling, or axis boundary conditions.

2 Conventions

The physical spatial dimension is

$$\text{GR_SPACEDIM} = 4,$$

and the gridded Chombo dimension is

$$\text{CH_SPACEDIM} = 2.$$

The gridded directions are x, z . The two hidden directions are equivalent w directions, so

$$N_{\text{hidden}} = 4 - 2 = 2.$$

The hidden component h_{ww} is Cartesian-like, not $g_{\theta\theta}$. The spherical/cartoon reconstruction supplies the external factor:

$$g_{\theta\theta} = x^2 \gamma_{ww} = x^2 \frac{h_{ww}}{\chi}.$$

All traces are four-dimensional, and the conformal extrinsic-curvature denominator remains $1/4$.

3 Cartoon Axis Versus Physical Singularity

In the reduced black-string cartoon setup, the cartoon/spherical axis is at $x = 0$. The unregularized black-string curvature singularity is also located at $x = 0$. These two issues are co-located in the coordinate description but are conceptually distinct.

The cartoon-axis problem is a removable coordinate-singularity, parity, and tensor-regularity problem for smooth $\text{SO}(3)$ -symmetric fields. The black-string singularity is a physical curvature singularity of the target spacetime. Turduckening or interior regularization is meant to remove or regularize the physical singularity from the active evolved problem. The cartoon-axis treatment separately handles the removable singular structures introduced by the spherical/cartoon reduction.

Therefore turduckening addresses the fact that the physical singularity is co-located with the cartoon axis, but it does not replace correct axis parity and removable-singularity treatment. Conversely, passing the small- x cartoon regularity checks does not prove that the black-string interior has been physically regularized correctly.

4 Apparent Singularities

Cartoon source terms near $x = 0$ contain apparent singular structures such as

$$\frac{1}{x}, \quad \frac{1}{x^2}, \quad \frac{\gamma_{xx} - \gamma_{ww}}{x^2}, \quad \frac{\partial_x q}{x}.$$

The corresponding conformal expressions involve h_{xx} , h_{ww} , χ , and their derivatives. These expressions can be removable singularities for smooth $\text{SO}(3)$ -symmetric data, but they are unsafe in raw numerical form unless the limiting behavior is enforced.

5 Parity And Matching

Scalar-like quantities such as χ , K , Θ , and the lapse are expected to be even in x . Diagonal components h_{xx} , h_{zz} , and h_{ww} are expected to be even. Components with one radial and one non-radial index, such as h_{xz} and A_{xz} , are expected to be odd. The radial connection and shift components, such as $\hat{\Gamma}^x$ and β^x , are expected to be odd, while the z -components are expected to be even. These component parities should be aligned with the chosen GRChombo-facing project conventions before coding.

The diagonal smooth-axis matching condition is

$$\gamma_{xx} - \gamma_{ww} = O(x^2).$$

Since $\gamma_{ij} = \chi^{-1}h_{ij}$, with χ finite and even at the axis, the corresponding conformal condition is

$$h_{xx} - h_{ww} = O(x^2).$$

For the off-diagonal component,

$$\gamma_{xz} = O(x), \quad h_{xz} = O(x), \quad A_{xz} = O(x).$$

6 Hatted Connection Guard

The highest-risk small-axis quantity is the hatted conformal connection, especially $\hat{\Gamma}^x$. Scalar-like variables and manifestly odd tensor fields can be checked with generic Taylor representatives. In contrast, $\hat{\Gamma}^x$ is assembled from contracted angular Christoffels that can carry bare $1/x$ behavior, and the hidden-sector multiplicity $N_{\text{hidden}} = 2$ is load-bearing.

The local Stage 3I symbolic guard checks the documented singular cartoon contribution

$$\tilde{\Gamma}_{\text{singular}}^x = \frac{N_{\text{hidden}}}{x} \left(1 - \frac{h_{xx}}{h_{ww}} \right), \quad N_{\text{hidden}} = 2.$$

With $h_{xx} - h_{ww} = O(x^2)$, this expression is finite and odd. If the axis values of h_{xx} and h_{ww} do not match, the same assembled expression exposes a residual $1/x$ divergence.

The hatted connection convention is

$$\hat{\Gamma}^i = \tilde{\Gamma}^i + 2Z^i.$$

The symbolic check assumes regular $Z^x = O(x)$ and verifies that $\hat{\Gamma}^x$ remains finite. The expected parity is that $\hat{\Gamma}^x$ is odd and $\hat{\Gamma}^z$ is even, but the final sign and full contracted-connection convention still need to be derived and documented in the GRChombo-facing cartoon extension.

A residual axis error in $\hat{\Gamma}^x$ can contaminate the Gamma-driver shift/gauge sector, so this assembled check is stronger than generic “odd-over- x ” tests.

7 Symbolic Limits

The Stage 3I SymPy check uses Taylor representatives

$$q(x, z) = q_0(z) + q_2(z)x^2 + O(x^4),$$

and

$$\gamma_{xz}(x, z) = s_1(z)x + s_3(z)x^3 + O(x^5).$$

It verifies limits such as

$$\lim_{x \rightarrow 0} \frac{\partial_x q}{x} = 2q_2(z),$$

and

$$\lim_{x \rightarrow 0} \frac{\gamma_{xx} - \gamma_{ww}}{x^2}$$

is finite when the axis matching condition is imposed. It also verifies that odd off-diagonal components have finite quotients such as

$$\lim_{x \rightarrow 0} \frac{\gamma_{xz}}{x} = s_1(z).$$

It also checks the assembled $\tilde{\Gamma}^x$ singular part and $\hat{\Gamma}^x = \tilde{\Gamma}^x + 2Z^x$ under regular Z^x parity. These checks validate local removable-singularity bookkeeping only; they do not derive the full CCZ4 RHS.

8 Deferred Implementation Choices

Stage 3I does not choose ghost-cell parity rules, one-sided stencils, axis-point evaluation, turduckening cutoff handling, finite-difference order, or C++ source-term organization. Those decisions are deferred to Stage 3J, Stage 3K, and the later implementation stage.